



Solo Leveling Real-Life Training System

Complete Blueprint & Research Reference

Version 2.0 · Designed for Beginners → Elite Athletes · Levels 1–100

1. VISION & PHILOSOPHY

This system is a real-life RPG modeled on the Solo Leveling manhwa — where the weakest hunter grinds relentlessly to become the strongest. The user starts at E-Rank (Level 1) and progresses toward S-Rank (Level 100), which represents the performance ceiling of an elite athlete. The system is brutalist: it does not care about feelings, mood, or comfort. It tracks only one thing — did you perform, or did you not?

Core Design Principles:

- No sympathy. Performance is the only variable that matters.
- Failure = hold the line. The system does not decrease difficulty when a user fails.
- Repeated failure = penalty escalation. Continuous non-completion triggers increased targets to shock the user out of stagnation.
- Success = immediate progression. Hit your target today, tomorrow's target goes up.
- Each category (Strength, Endurance, Martial Arts) has its own independent progression curve.
- The system is for anyone — a school student who has never exercised, or an existing athlete.

2. RANK & LEVEL STRUCTURE

Rank	Level Range	Description	Example Benchmark
E	1 – 20	Complete beginner, sedentary lifestyle	5 push-ups, 1 km run/walk
D	21 – 40	Building baseline fitness, consistent habits	25 push-ups, 3 km run
C	41 – 60	Intermediate fitness, real strength forming	50 push-ups, 5 km run
B	61 – 75	Advanced athlete, high discipline	80 push-ups, 8 km run
A	76 – 90	Near-elite, exceptional strength & endurance	100+ push-ups, 10 km run
S	91 – 100	Elite athlete tier — conceptual ceiling	One-arm push-ups, 42 km run

Level 100 is the conceptual ceiling of elite athleticism — not a hard stop. Once a user reaches Level 100, the system shifts from prescriptive to tracker-only mode. The user self-directs, and the system logs performance without enforcing targets. Levels 101+ exist but carry no defined benchmarks.

3. BASELINE ASSESSMENT PROTOCOL

Research Basis: A standardized max-rep test to fatigue is the most validated field method for assessing muscular endurance in general populations. (ACSM Guidelines for Exercise Testing and Prescription, 11th Ed.; Canadian Society for Exercise Physiology — CSEP Physical Activity Training for Health).

3.1 Assessment Structure

- Test must be performed fresh — not after any other exercise. Full recovery between movement tests (minimum 5 minutes).
- Each test: perform max reps of a single movement to genuine fatigue with no rest mid-set.
- Record: total reps completed, time taken, perceived difficulty (RPE 1–10 scale).

3.2 Assessment Battery

Movement	Test Protocol	Measures
Push-Up	Max reps to fatigue, no rest	Upper body pushing strength + endurance
Bodyweight Squat	Max reps to fatigue, no rest	Lower body strength + endurance
Plank Hold	Hold until form breaks (seconds)	Core endurance
1-Mile Run/Walk	Complete 1 mile, record time	Cardiovascular baseline
Jab (Boxing)	50 reps, observe form quality	Coordination & martial baseline

Result: The baseline score for each category sets the Day 1 target at 80% of the user's tested max. This ensures the first session is achievable but not trivial — a critical condition for long-term adherence (Locke & Latham Goal-Setting Theory, 2002).

4. PROGRESSION MATHEMATICS

4.1 Core Principle — Progressive Overload

The foundational research consensus (Kraemer & Ratamess, 2004; NSCA Essentials of Strength Training) states that sustainable strength and endurance gains require a 5–10% weekly increase in training volume. Going above 10% per week significantly raises injury risk (the '10% Rule,' widely validated in running literature and general exercise science).

4.2 Daily Progression Logic

Since the user progresses every single day (not weekly), we distribute the 5–10% weekly gain across 7 days:

Daily multiplier = $(1 + \text{weekly_rate})^{1/7}$

At 7% weekly: daily multiplier = $1.07^{1/7} \approx 1.0097$ (~+1% per day)

At 10% weekly: daily multiplier = $1.10^{1/7} \approx 1.0138$ (~+1.4% per day)

The system uses a tiered daily rate based on rank, since advanced athletes adapt more slowly:

Rank	Level Range	Weekly Volume Target	Daily Multiplier	Rationale
E	1–20	+8% / week	×1.011	Beginner adaptation is fastest; neural gains dominate
D	21–40	+7% / week	×1.010	Structural adaptation phase begins
C	41–60	+5% / week	×1.007	Intermediate plateau; slower but consistent
B	61–75	+4% / week	×1.006	Advanced — volume increases must be precise
A/S	76–100	+2–3% / week	×1.003–1.004	Elite plateau management; micro-progression

4.3 Failure & Penalty Escalation Logic

Standard rule: If user fails to hit today's target → tomorrow's target stays the same.

Penalty escalation rule (consecutive failures):

- Fail 1–2 consecutive days → target held at same value.
- Fail 3 consecutive days → target increases by +5% ('shock escalation'). System does not wait.
- Fail 5+ consecutive days → target increases by +10%. The system interprets prolonged failure as avoidance, not incapacity.

Psychological basis: Extinction burst theory and operant conditioning research (Skinner, 1938; modern athletic psychology) support the principle that removing all pressure on failure reinforces avoidance behavior. Escalation under failure mimics the psychological effect of consequence-based motivation.

5. EXERCISE CATEGORY PROGRESSIONS

5.1 Strength (Push-Based)

- Push-Up Progression Ladder: Standard Push-Up → Diamond Push-Up → Wide Push-Up → Archer Push-Up → Decline Push-Up → Pike Push-Up → One-Arm Negative → One-Arm Push-Up
- Transition Rule: When user consistently completes 3 sets of the current exercise variation at the prescribed rep target for 3+ consecutive days, the system flags a variation upgrade.
- Volume conversion on upgrade: New exercise target = $\text{floor}(\text{old_reps} \times 0.50)$ distributed across sets. Example: 30 standard push-ups → 2 × 15 archer push-ups.

Research: Archer push-ups shift ~60-70% of load onto the working arm vs. ~35% per arm in standard push-ups. A 50% rep reduction maintains comparable absolute loading per limb. (Gouvali & Boudolos, 2005 — EMG analysis of push-up variations.)

5.2 Endurance (Cardio-Based)

- Run Progression: Walk/Run intervals → Continuous 1km → 3km → 5km → 10km → Half-Marathon pace → Marathon pace
- Distance increase: +10% per week maximum (the validated '10% Rule' for running: van Gent et al., 2007, British Journal of Sports Medicine).
- Pace work begins at C-Rank: once distance milestones are hit, the system introduces time targets for the same distance.
- Plank and isometric holds: +5 seconds per successful day, with a set-split when holds exceed 90 seconds (split into 2 × 45s).

5.3 Martial Arts (Skill-Based)

- Assumption: User starts with zero martial arts experience. No baseline test. System prescribes rep targets directly.
- Move introduction: One new move introduced every 7–14 days based on progression through a curated library.
- Learning protocol: Day 1–2 = practice reps at reduced volume (40% of target) for form learning. Day 3+ = full target volume.
- Rep tracking: User self-reports completion. No form validation. System trusts user input.
- Daily target: Starts at 50 reps of the assigned move. Increases by 10 reps every 3 days of successful completion.

Phase	Moves Introduced
E-Rank	Jab, Cross, Roundhouse Kick, Front Kick, Guard Stance
D-Rank	Hook, Uppercut, Side Kick, Elbow Strike, Clinch Defense
C-Rank	Jab-Cross Combo, Switch Kick, Knee Strike, Overhand, Teep
B-Rank	Spinning Back Kick, Axe Kick, Body Hook, Double Jab, Catch & Counter
A-Rank	Spinning Heel Kick, Flying Knee, Combination Flows (4+ strike sequences)
S-Rank	Full combination drills, shadow boxing rounds, advanced footwork patterns

6. PLATEAU DETECTION & DYNAMIC EXERCISE SWAPS

A plateau is defined as: the user successfully hitting the same target rep/set count for 3+ consecutive days without the system having increased it (i.e. they've maxed out the variation, not failed it). This is distinct from failure.

6.1 Detection Algorithm

For each exercise variation, the system tracks: `consecutive_success_days`. When `consecutive_success_days` ≥ 3 AND the user is performing the top set configuration for that variation (e.g. 3 sets), the system triggers a variation swap on the next session.

6.2 Swap Volume Formula

`new_target_reps = floor(current_reps × difficulty_drop_factor)`

`new_sets = max(1, current_sets - 1)`

- Standard variation upgrade (e.g. standard → diamond): drop factor = 0.70, same sets
- Major variation upgrade (e.g. standard → archer): drop factor = 0.50, split sets
- New movement category (e.g. push-up → handstand pushup progression): drop factor = 0.40, 1 set

This formula ensures total mechanical work (sets × reps) stays within 70–100% of prior session volume — maintaining stimulus without overload on an unfamiliar movement pattern. (Principle: Specificity of Adaptation, NSCA, 2016.)

7. XP & LEVEL SYSTEM

7.1 XP Curve — Exponential Scaling

XP required to reach Level N uses an exponential curve so early levels are achievable quickly (sustaining motivation) while higher levels require compounding effort:

$XP_{required}(N) = base_xp \times growth_rate^{(N-1)}$

`base_xp = 100 XP (Level 1 → 2)` `growth_rate = 1.15`

Level 10 requires: $100 \times 1.15^9 \approx 352$ XP

Level 50 requires: $100 \times 1.15^{49} \approx 107,000$ XP

Level 100 requires: $100 \times 1.15^{99} \approx 1.17$ billion XP (elite ceiling)

This exponential curve means Level 100 is mathematically approachable only through years of consistent elite performance — exactly the intended design.

7.2 XP Award Logic

- Complete a daily quest: `Base XP = difficulty_multiplier × reps_completed`
- Hit exact target: +20% XP bonus
- Exceed target: +50% XP bonus (system still uses target as next-day baseline, not the exceeded amount)
- Streak bonus: +5% XP per consecutive day, up to +50% cap at 10-day streak
- New variation mastered: +200 XP one-time bonus
- Rank-up milestone: +500 XP flat bonus

8. RESEARCH REFERENCES

Progressive Overload & Strength Adaptation

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Fitness Testing Battery

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Specificity of Adaptation

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Operant Conditioning & Behavioral Reinforcement

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9. DEVELOPMENT ROADMAP

Phase 1 — Core Engine (V2)

- Baseline assessment flow on first launch
- Independent progression curves per category (Strength, Endurance, Martial Arts)
- Daily target calculation using rank-tiered multipliers
- Failure tracking with penalty escalation logic
- Exercise variation library with swap triggers
- Exponential XP curve and rank system E → S

Phase 2 — Intelligence Layer

- Plateau detection algorithm per exercise variation
- Dynamic swap engine with volume conversion formula
- Martial arts move library with YouTube link integration
- Streak system with XP bonuses

Phase 3 — Polish & Persistence

- Local storage save/load (all progress persists)
- Animated rank-up and level-up screens
- Solo Leveling dark theme UI — purple/black/gold
- Daily quest card system with completion animation
- Progress visualization: charts per category over time

ARISE. FROM E-RANK TO S-RANK. THE SYSTEM DOES NOT WAIT.

Solo Leveling Training System Blueprint — Internal Reference Document